SAT Math Mastery · Session 5

Equivalent Expressions and Exponential Functions

Time 3 hours: Part A in class (about 2 hours of solving), then Part B, the self-assessment, done independently (about 50 minutes), and a short error log.

Topics equivalent expressions: exponent rules, rational expressions, factoring and identities · radical equations · systems with exponents and radicals · radical functions and domain · exponential growth and decay models · interpreting exponential models · recursive and geometric sequences · successive percent changes

Name / Date _____ _____

PART A · IN CLASS — 35 PROBLEMS

Each problem states its tool. Write your work in the space below. Multiple choice: one correct option. Student-produced response (SPR): enter an exact value (fraction or decimal). Grey boxes show exactly what to type in Desmos or on your calculator, one expression per line; tap a grey dot to read an intersection, intercept or vertex. A letter such as a, c or k becomes a slider when you accept Desmos's offer to add one.

1. HARD · NO CALCULATOR NEEDED

For all real values of x,

$$\frac{6^{2x+3}}{2^{2x-1}} = k \cdot 9^x$$

where k is a constant. What is the value of k?

1. Split the base: $6^{2x+3} = 2^{2x+3} \cdot 3^{2x+3}$.
2. The powers of 2 cancel down: $\frac{2^{2x+3}}{2^{2x-1}} = 2^{(2x+3)-(2x-1)} = 2^4 = 16$.
3. The powers of 3: $3^{2x+3} = 3^3 \cdot (3^2)^x = 27 \cdot 9^x$.
4. So the left side is $16 \cdot 27 \cdot 9^x = 432 \cdot 9^x$ and $k = 432$.

Answer: 432

2. HARD · NO CALCULATOR NEEDED

$$(x+p)(x+q) = x^2 + bx + c$$

In the equation above, p and q are real constants and b and c are integers. Which of the following must be true?

I. $p+q$ is an integer

II. p^2+q^2 is an integer

III. $\frac{p^2+q^2}{2}$ is an integer

(A) I only

(B) I and II only ✓

(C) II and III only

(D) I, II, and III

1. Expand: $(x+p)(x+q) = x^2 + (p+q)x + pq$, so $b = p+q$ and $c = pq$, both integers. Statement I is true.
2. $p^2+q^2 = (p+q)^2 - 2pq = b^2 - 2c$, an integer, so statement II is true.
3. Statement III needs $\frac{b^2-2c}{2}$ to be an integer, which fails when b is odd. Try $p=1$, $q=2$ ($b=3$, $c=2$): $p^2+q^2 = 5$, and $\frac{5}{2}$ is not an integer.
4. Only I and II must be true: option (B).

Answer: (B)

3. HARD · NO CALCULATOR NEEDED

For all $x > 1$,

$$\frac{a}{x-1} - \frac{bx+c}{x^2+x+1} = \frac{3x^2+8x+4}{x^3-1}$$

where a, b, and c are constants. What is the value of c?

1. Multiply both sides by $x^3 - 1 = (x-1)(x^2+x+1)$: $a(x^2+x+1) - (bx+c)(x-1) = 3x^2+8x+4$.

2. Put $x = 1$ so that the second term vanishes: $3a = 3+8+4 = 15$, so $a = 5$.

3. Put $x = 0$: $a + c = 4$, so $c = 4 - 5 = -1$.

4. Check with the x^2 and x coefficients: $5 - b = 3$ gives $b = 2$, and $5 + b - c = 5 + 2 + 1 = 8$ as required.

Answer: -1

4. HARD · NO CALCULATOR NEEDED

For $y \neq 0$ and $y \neq \dfrac{1}{3}$, which of the following is equivalent to

$$\frac{\dfrac{1}{y^3} - 27}{\dfrac{1}{y} - 3} \; ?$$

(A) $\dfrac{1}{y^2} - \dfrac{3}{y} + 9$

(B) $\dfrac{1}{y^2} + 9$

(C) $\dfrac{1}{y^2} + \dfrac{6}{y} + 9$

(D) $\dfrac{1}{y^2} + \dfrac{3}{y} + 9$ ✓

1. Let $u = \dfrac{1}{y}$. The expression becomes $\dfrac{u^3 - 27}{u - 3}$.

2. Difference of cubes: $u^3 - 27 = (u-3)(u^2+3u+9)$ (the middle sign is $+$). Check: $(u-3)(u^2+3u+9) = u^3+3u^2+9u-3u^2-9u-27 = u^3-27$.

3. Cancel $u - 3$ (not zero because $y \neq \frac{1}{3}$): $u^2 + 3u + 9 = \dfrac{1}{y^2} + \dfrac{3}{y} + 9$, option (D).

Answer: (D)

5. HARD · NO CALCULATOR NEEDED

The polynomial $p(x) = x^3 - 9x^2 + kx - 12$ has a positive integer constant k. For some positive integer n, $x - n$ is a factor of $p(x)$. What is the sum of all possible values of k?

1. If $x - n$ is a factor then $p(n) = 0$: $n^3 - 9n^2 + kn - 12 = 0$, so $k = \dfrac{12}{n} + 9n - n^2$. For k to be an integer, n must divide 12: $n \in \{1, 2, 3, 4, 6, 12\}$.

2. Compute k for each: $n = 1$: $k = 20$; $n = 2$: $k = 20$; $n = 3$: $k = 22$; $n = 4$: $k = 23$; $n = 6$: $k = 20$; $n = 12$: $k = -35$.

3. Keep only positive k and count each value once: $k \in \{20, 22, 23\}$ ($n = 12$ gives a negative k).

4. The sum is $20 + 22 + 23 = 65$.

Answer: 65

6. HARD · NO CALCULATOR NEEDED

$$P = \frac{a}{\sqrt[5]{w^2}}$$

The formula above relates the positive quantities P and w, where a is a positive constant. Which of the following expresses w in terms of P and a?

(A) $w = \left(\frac{a}{P}\right)^{5/2}$ ✓

(B) $w = \left(\frac{P}{a}\right)^{5/2}$

(C) $w = \left(\frac{a}{P}\right)^{2/5}$

(D) $w = \left(\frac{P}{a}\right)^{2/5}$

1. Rewrite the radical as a power: $\sqrt[5]{w^2} = w^{2/5}$, so $P = \frac{a}{w^{2/5}}$.

2. Multiply by $w^{2/5}$ and divide by P: $w^{2/5} = \frac{a}{P}$ (the a stays on top).

3. Raise both sides to the reciprocal power $\frac{5}{2}$: $w = \left(\frac{a}{P}\right)^{5/2}$, option (A). Options with the exponent $\frac{2}{5}$ forgot to flip it.

Answer: (A)

7. HARD · NO CALCULATOR NEEDED

For all real numbers x,

$$8^x \cdot 4^{x/3} \cdot \left(\sqrt{2}\right)^x = b^x,$$

where b is a positive constant. Which of the following is equal to 2?

(A) $b^{6/25}$ ✓

(B) $b^{25/6}$

(C) b^{25}

(D) $b^{-6/25}$

1. Write every base as a power of 2: $8^x = 2^{3x}$, $4^{x/3} = 2^{2x/3}$, $(\sqrt{2})^x = 2^{x/2}$.

2. Add the exponents: $3 + \frac{2}{3} + \frac{1}{2} = \frac{25}{6}$, so $b^x = 2^{\frac{25}{6}x}$ and $b = 2^{\frac{25}{6}}$.

3. Raise b to the reciprocal power $\frac{6}{25}$: $b^{\frac{6}{25}} = 2$, option (A).

Answer: (A)

8. HARD · NO CALCULATOR NEEDED

$$\frac{2}{a} - \frac{3}{b} = \frac{5}{c}$$

In the equation above, a, b and c are nonzero constants and $2c \neq 5a$. Which of the following expresses b in terms of a and c?

(A) $b = \dfrac{3ac}{5a - 2c}$

(B) $b = \dfrac{2c - 5a}{3ac}$

(C) $b = \dfrac{3ac}{2c + 5a}$

(D) $b = \dfrac{3ac}{2c - 5a}$ ✓

1. Get the b term alone: $\dfrac{3}{b} = \dfrac{2}{a} - \dfrac{5}{c}$.
2. Common denominator ac: $\dfrac{3}{b} = \dfrac{2c - 5a}{ac}$.
3. Take the reciprocal of both sides and multiply by 3: $b = \dfrac{3ac}{2c - 5a}$, option (D). The condition $2c \neq 5a$ is what keeps this denominator nonzero.

Answer: (D)

9. HARD · NO CALCULATOR NEEDED

The polynomial $6x^4 + kx^2 + 24$, where k is a constant, can be written as $(2x^2 + a)(3x^2 + b)$, where a and b are integers. What is the greatest possible value of k?

1. Expand: $(2x^2 + a)(3x^2 + b) = 6x^4 + (2b + 3a)x^2 + ab$, so $ab = 24$ and $k = 2b + 3a$.
2. List the positive pairs with $ab = 24$ (negative pairs give negative k): $(1, 24) \to k = 51$; $(2, 12) \to k = 30$; $(3, 8) \to k = 25$; $(4, 6) \to k = 24$; $(6, 4) \to k = 26$; $(8, 3) \to k = 30$; $(12, 2) \to k = 40$; $(24, 1) \to k = 74$.
3. The greatest value is $k = 74$ (from $a = 24$, $b = 1$).

Answer: 74

10. HARD · NO CALCULATOR NEEDED

For a real number a with $|a| \neq 1$,

$$\frac{1}{1-a} + \frac{1}{1+a} + \frac{2}{1+a^2} + \frac{4}{1+a^4} = 10.$$

What is the value of a^8?

(A) $\dfrac{5}{4}$

(B) $\dfrac{9}{5}$

(C) $\dfrac{1}{5}$ ✓

(D) $\dfrac{4}{5}$

1. Add the first two fractions: $\dfrac{1}{1-a} + \dfrac{1}{1+a} = \dfrac{2}{1-a^2}$.
2. Add $\dfrac{2}{1+a^2}$: $\dfrac{2(1+a^2) + 2(1-a^2)}{1-a^4} = \dfrac{4}{1-a^4}$. Add $\dfrac{4}{1+a^4}$ the same way: $\dfrac{8}{1-a^8}$.
3. Solve $\dfrac{8}{1-a^8} = 10$: $1 - a^8 = \dfrac{8}{10}$, so $a^8 = 1 - \dfrac{4}{5} = \dfrac{1}{5}$, option (C).

Answer: (C)

11. HARD · NO CALCULATOR NEEDED

For real numbers a and b, $a^2 - b^2 = 36$. What is the value of $(3a + b)^2 - (a + 3b)^2$?

1. Expand: $(3a + b)^2 = 9a^2 + 6ab + b^2$ and $(a + 3b)^2 = a^2 + 6ab + 9b^2$.

2. Subtract: the $6ab$ terms cancel, leaving $8a^2 - 8b^2 = 8(a^2 - b^2)$.

3. Substitute $a^2 - b^2 = 36$: $8 \cdot 36 = 288$.

Answer: 288

12. HARD · NO CALCULATOR NEEDED

If $2x + 3y = 8$, which of the following is equivalent to $2x^2 + 3xy - 4x - 6y$?

(A) $8x - 8$ (B) $16 - 8x$

(C) $8x - 4y$ **(D)** $8x - 16$ ✓

1. Factor by grouping: $2x^2 + 3xy - 4x - 6y = x(2x + 3y) - 2(2x + 3y) = (x - 2)(2x + 3y)$.

2. Substitute $2x + 3y = 8$: $8(x - 2) = 8x - 16$, option (D).

3. Quick check: $x = 1$ gives $y = 2$, and the original expression is -8, while $8x - 16 = -8$.

Answer: (D)

13. HARD · NO CALCULATOR NEEDED

Let t be a positive real number with $t \neq 1$. Define $m = \sqrt[3]{t^2}$ and $n = \sqrt{t}$. If

$$\frac{m^p \cdot n^3}{\sqrt[6]{t}} = t^5,$$

what is the value of p?

1. Write everything as powers of t: $m = t^{2/3}$ and $n = t^{1/2}$, so $m^p = t^{2p/3}$ and $n^3 = t^{3/2}$.

2. Combine the exponents on the left: $\frac{2p}{3} + \frac{3}{2} - \frac{1}{6} = \frac{2p}{3} + \frac{4}{3}$.

3. Equate exponents ($t \neq 1$): $\frac{2p}{3} + \frac{4}{3} = 5$, so $\frac{2p}{3} = \frac{11}{3}$ and $p = \frac{11}{2}$ (which is 5.5 as a decimal).

Answer: $\frac{11}{2}$ **or** 5.5

14. HARD · NO CALCULATOR NEEDED

The expression $\dfrac{(2x-3)^4(x+2)-7}{5}$ is equivalent to $ax^5+bx^4+cx^3+dx^2+ex+f$, where a, b, c, d, e and f are constants. What is the value of $a+b+c+d+e+f$?

(A) $-\dfrac{4}{5}$ ✓ (B) $\dfrac{3}{5}$

(C) -4 (D) 31

1. The sum $a+b+c+d+e+f$ is the value of the polynomial at $x=1$ (every power of 1 is 1), so no expanding is needed.
2. At $x=1$: $(2-3)^4(1+2)=1\cdot 3=3$.
3. Then $\dfrac{3-7}{5}=-\dfrac{4}{5}$, option (A).

Answer: (A)

15. HARD · NO CALCULATOR NEEDED

The polynomial $12x^4-84x^2+144$ can be written in the form $(ax^2+b)(cx^2+d)$, where a, b, c and d are integers with $a>0$ and $c>0$. What is the least possible value of $a+b+c+d$?

(A) -27 (B) -22

(C) -18 **(D)** -38 ✓

1. Factor out 12 and then the quadratic in x^2: $12x^4-84x^2+144=12(x^2-3)(x^2-4)$.
2. Split the 12 as ac: $(ax^2-3a)(cx^2-4c)$ with $ac=12$. Then $a+b+c+d=a+c-3a-4c=-2a-3c$. (Swapping the roles of the two factors gives $-3a-2c$.)
3. To make this as small as possible, put the big coefficient on the larger multiplier: $a=1$, $c=12$ gives $-2(1)-3(12)=-38$, from $(x^2-3)(12x^2-48)$ with $1-3+12-48=-38$.
4. The other splits give sums such as -27, -22, -18 (these are the other options), all larger, so the least value is -38, option (D).

Answer: (D)

16. HARD · NO CALCULATOR NEEDED

The function f is defined by $f(x)=\dfrac{6^{x+1}\cdot 4^x}{3^{x-2}}$. It can be written in the form $f(x)=a\cdot b^{x-1}$, where a and b are positive constants. What is the value of $a+b$?

(A) 62 **(B)** 440 ✓

(C) 456 (D) 3456

1. Write the bases with primes: $6^{x+1}=2^{x+1}3^{x+1}$ and $4^x=2^{2x}$.
2. Divide by 3^{x-2}: the powers of 3 give $3^{(x+1)-(x-2)}=3^3=27$; the powers of 2 give $2^{3x+1}=2\cdot 8^x$. So $f(x)=54\cdot 8^x$.
3. Match the form $a\cdot b^{x-1}$: $b=8$ and $54\cdot 8^x=54\cdot 8\cdot 8^{x-1}=432\cdot 8^{x-1}$, so $a=432$.
4. Then $a+b=432+8=440$, option (B).

Answer: (B)

17. HARD · NO CALCULATOR NEEDED

For a positive real number m, let $a = \sqrt[3]{m}$ and $b = \sqrt[4]{m}$. If $ab = 128$, what is the value of ab^2?

1. $ab = m^{1/3} \cdot m^{1/4} = m^{\frac{7}{12}}$.

2. Since $128 = 2^7$: $m^{\frac{7}{12}} = 2^7$, so $m = 2^{7 \cdot \frac{12}{7}} = 2^{12}$.

3. Now $ab^2 = m^{1/3} \cdot m^{2/4} = m^{\frac{5}{6}} = 2^{12 \cdot \frac{5}{6}} = 2^{10} = 1024$.

Answer: 1,024

18. HARD · NO CALCULATOR NEEDED

For all real numbers x, the function g is defined by $g(x) = \dfrac{9^{x+2} \cdot 10^{x+1}}{6^{x-1}}$. The function can be written in the form $g(x) = k \cdot 15^{x+1}$, where k is a constant. What is the value of k?

1. Use prime bases: $9^{x+2} = 3^{2x+4}$, $10^{x+1} = 2^{x+1}5^{x+1}$, $6^{x-1} = 2^{x-1}3^{x-1}$.

2. Divide: powers of 2 give $2^{(x+1)-(x-1)} = 2^2 = 4$; powers of 3 give $3^{(2x+4)-(x-1)} = 3^{x+5}$; and 5^{x+1} stays.

3. Pull out 3^4 so the other 3 and 5 pair up into 15: $3^{x+5} \cdot 5^{x+1} = 3^4 \cdot (3 \cdot 5)^{x+1} = 81 \cdot 15^{x+1}$.

4. So $g(x) = 4 \cdot 81 \cdot 15^{x+1}$ and $k = 324$.

Answer: 324

19. HARD · NO CALCULATOR NEEDED

Positive real numbers a and b, both greater than 1, satisfy $a^3 = b^2$. If $\dfrac{a^9}{\sqrt[3]{b}} = b^k$, what is the value of k?

(A) $\dfrac{17}{3}$ ✓ (B) 6

(C) $\dfrac{19}{3}$ (D) $\dfrac{17}{2}$

1. From $a^3 = b^2$ we get $a = b^{2/3}$.

2. Then $a^9 = b^{\frac{2}{3} \cdot 9} = b^6$.

3. Divide by $\sqrt[3]{b} = b^{1/3}$: $b^{6-1/3} = b^{\frac{17}{3}}$, so $k = \dfrac{17}{3}$, option (A).

Answer: (A)

20. HARD · NO CALCULATOR NEEDED

Positive real numbers x and y satisfy $x \neq y$, $x + y = 15$ and $xy = 36$. What is the value of

$$\frac{\dfrac{1}{x} - \dfrac{1}{y}}{\dfrac{1}{x^2} - \dfrac{1}{y^2}} \; ?$$

(A) $\dfrac{5}{12}$ (B) 1

(C) 4 **(D) $\dfrac{12}{5}$ ✓**

1. Factor the bottom as a difference of squares: $\dfrac{1}{x^2} - \dfrac{1}{y^2} = \left(\dfrac{1}{x} - \dfrac{1}{y}\right)\left(\dfrac{1}{x} + \dfrac{1}{y}\right)$.

2. Cancel $\dfrac{1}{x} - \dfrac{1}{y}$ (not zero since $x \neq y$): the expression is $\dfrac{1}{\frac{1}{x} + \frac{1}{y}} = \dfrac{xy}{x + y}$.

3. Substitute: $\dfrac{36}{15} = \dfrac{12}{5}$, option (D).

Answer: (D)

21. HARD · DESMOS

What is the solution set of the equation

$$\sqrt{x + 7} = x - 5?$$

(A) $x = 2$ only **(B) $x = 9$ only ✓**

(C) $x = 2$ and $x = 9$ (D) There is no solution.

1. Graph each side: type `y=sqrt(x+7)` and `y=x-5`.

2. Tap the intersection: (9, 4). The line and the curve meet only once, so there is exactly one solution, $x = 9$.

3. Type `sqrt(9+7)`: it returns 4; type `9-5`: it returns 4. Both sides agree at $x = 9$.

4. Squaring also produces $x = 2$, but at $x = 2$ the line is at -3 while the curve is at 3: extraneous. The answer is option (B).

Answer: (B)

22. HARD · DESMOS

a and b are positive consecutive integers, and $b > a$. If b is thirteen more than $\sqrt{a}$, what is the value of a?

1. Consecutive integers with $b > a$ means $b = a + 1$, so $a + 1 = 13 + \sqrt{a}$. Type `y=x+1` and `y=13+sqrt(x)` (letting x stand for a).

2. Tap the intersection: (16, 17), so $a = 16$ (the curves cross only once).

3. Type `13+sqrt(16)`: it returns 17, which equals $16 + 1$: confirmed, with $b = 17$.

Answer: 16

23. HARD · NO CALCULATOR NEEDED

Let m be an even integer. How many possible values of m satisfy $\sqrt{m+11} \le 4$?

1. The square root needs $m + 11 \ge 0$, so $m \ge -11$.

2. Square both sides of $\sqrt{m+11} \le 4$ (both sides are non-negative): $m + 11 \le 16$, so $m \le 5$.

3. The even integers from -11 to 5 are $-10, -8, -6, -4, -2, 0, 2, 4$, a total of 8 values.

Answer: 8

24. HARD · NO CALCULATOR NEEDED

Real numbers x and y satisfy

$$\sqrt[3]{2^x} \cdot 4^y = 256 \qquad \text{and} \qquad \frac{16^x}{\sqrt[3]{4^y}} = 2^{22}.$$

What is the value of xy?

1. Use base 2: $\sqrt[3]{2^x} \cdot 4^y = 2^{x/3+2y} = 2^8$, so $\dfrac{x}{3} + 2y = 8$, i.e. $x + 6y = 24$.

2. Second equation: $\dfrac{16^x}{\sqrt[3]{4^y}} = 2^{4x-2y/3} = 2^{22}$, so $4x - \dfrac{2y}{3} = 22$, i.e. $12x - 2y = 66$, i.e. $6x - y = 33$.

3. From $6x - y = 33$: $y = 6x - 33$. Substitute: $x + 6(6x - 33) = 24$, so $37x = 222$ and $x = 6$, $y = 3$.

4. Then $xy = 6 \cdot 3 = 18$.

Answer: 18

25. HARD · NO CALCULATOR NEEDED

Positive real numbers x and y satisfy

$$\sqrt{x} - \sqrt{y} = 2 \qquad \text{and} \qquad x + y = 26.$$

What is the value of xy?

1. Square the first equation: $x - 2\sqrt{xy} + y = 4$.

2. Substitute $x + y = 26$: $26 - 2\sqrt{xy} = 4$, so $\sqrt{xy} = 11$.

3. Square again: $xy = 11^2 = 121$.

Answer: 121

26. HARD · NO CALCULATOR NEEDED

The function $m(z) = -\sqrt{z^2 + pz + q}$ has graph passing through $(6, 0)$ and $\left(0, -\sqrt{252}\right)$, where p and q are constants. What is the greatest value of t for which $m(t) = 0$?

1. A zero of m is a zero of $z^2 + pz + q$. The point $(0, -\sqrt{252})$ gives $q = 252$.

2. The point $(6, 0)$ gives $36 + 6p + 252 = 0$, so $p = -48$.

3. Solve $z^2 - 48z + 252 = 0$: $(z - 6)(z - 42) = 0$ since $6 \cdot 42 = 252$ and $6 + 42 = 48$. The zeros are 6 and 42, so the greatest is 42.

Answer: 42

27. HARD · NO CALCULATOR NEEDED

The function $h(x) = -\sqrt{x^2 + bx + c}$ has a graph passing through $(2, 0)$ and $\left(0, -\sqrt{314}\right)$, where b and c are constants. What is the distance between the two x-intercepts of the graph?

1. The point $(0, -\sqrt{314})$ gives $c = 314$. The point $(2, 0)$ gives $4 + 2b + 314 = 0$, so $b = -159$.

2. The x-intercepts are the zeros of $x^2 - 159x + 314$, which factors as $(x - 2)(x - 157)$ because $2 \cdot 157 = 314$ and $2 + 157 = 159$.

3. The intercepts are $x = 2$ and $x = 157$, so the distance between them is $157 - 2 = 155$.

Answer: 155

28. HARD · NO CALCULATOR NEEDED

The function $f(x) = 2\left(3x + 1 - \sqrt{5}\right)\left(3x + 1 + \sqrt{5}\right)$ can be written as $f(x) = ax^2 + bx + c$, where a, b and c are constants. What is the value of $\dfrac{b + c}{a}$?

1. Spot the difference of squares with $u = 3x + 1$: $(u - \sqrt{5})(u + \sqrt{5}) = u^2 - 5$.

2. So $f(x) = 2\left((3x + 1)^2 - 5\right) = 2(9x^2 + 6x - 4) = 18x^2 + 12x - 8$.

3. Read off $a = 18$, $b = 12$, $c = -8$: $\dfrac{b + c}{a} = \dfrac{4}{18} = \dfrac{2}{9}$.

Answer: $\dfrac{2}{9}$ **or** 0.2222

29. Hard · Desmos

The polynomial function p is defined by $p(x) = -(x+3)^2(x-1)^3(x-4)^2$. Consider the following statements.
I. $p(x) \geq 0$ for all $x < 1$.
II. The graph of $y = p(x)$ crosses the x-axis at $x = 4$.
III. $p(x) \to \infty$ as $x \to -\infty$.
Which of the statements must be true?

(A) I only

(B) I and II only

(C) I and III only ✓

(D) II and III only

1. Type `y=-(x+3)^2(x-1)^3(x-4)^2` and zoom out. The graph meets the x-axis at $x = -3$, $x = 1$ and $x = 4$ (tap them to see).
2. Statement I: to the left of $x = 1$ the graph stays on or above the x-axis (it only touches it at $x = -3$). True.
3. Statement II: at $x = 4$ the graph touches the axis and turns back (even power), it does not cross. False.
4. Statement III: far to the left the graph rises (odd degree, negative leading coefficient). True. So I and III only: option (C).

Answer: (C)

30. Hard · No calculator needed

The function h is defined by $h(x) = \sqrt{x-2} + 1$. The function g is defined by $g(x) = h(x+5)$. What is the domain of g?

(A) $x \geq -3$ ✓

(B) $x \geq 2$

(C) $x \geq 7$

(D) $x \geq -5$

1. Substitute into h: $g(x) = h(x+5) = \sqrt{(x+5)-2} + 1 = \sqrt{x+3} + 1$.
2. The radicand must be non-negative: $x + 3 \geq 0$, so $x \geq -3$.
3. Shortcut: replacing x by $x + 5$ shifts the graph 5 units left, so the domain edge moves from 2 to $2 - 5 = -3$, option (A).

Answer: (A)

31. Hard · No calculator needed

A sequence is defined recursively by $a_1 = 5$ and $a_n = 2a_{n-1} - 3$ for $n \geq 2$. What is the value of a_4?

1. Apply the rule to the previous term each time, starting from $a_1 = 5$.
2. $a_2 = 2(5) - 3 = 7$ and $a_3 = 2(7) - 3 = 11$.
3. $a_4 = 2(11) - 3 = 19$.

Answer: 19

32. HARD · NO CALCULATOR NEEDED

A water-treatment filter reduces the concentration of a contaminant by 12% every hour it operates. If the initial concentration is 250 parts per million (ppm), which of the following functions models the concentration $C(t)$, in ppm, after t hours of operation?

(A) $C(t) = 250(1.12)^t$

(B) $C(t) = 250 - 12t$

(C) $C(t) = 250(0.12)^t$

(D) $C(t) = 250(0.88)^t$ ✓

1. A 12% decrease each hour leaves $100\% - 12\% = 88\%$, so the hourly factor is 0.88.

2. That gives $C(t) = 250(0.88)^t$, option (D).

3. Check at $t = 1$: $250(0.88) = 220$, which is $250 - 12\%$ of $250 = 250 - 30$.

Answer: (D)

33. HARD · NO CALCULATOR NEEDED

In a sequence, the first term is -4. Each term after the first is two less than triple the preceding term. Which term is the first one whose absolute value is greater than 150?

(A) 3

(B) 4

(C) 5 ✓

(D) 6

1. "Two less than triple the preceding term" means $a_{n+1} = 3a_n - 2$, starting from $a_1 = -4$.

2. Build the terms: $a_1 = -4$, $a_2 = -14$, $a_3 = -44$, $a_4 = -134$, $a_5 = -404$.

3. $|a_4| = 134$ is still below 150, and $|a_5| = 404$ is the first to exceed it: option (C).

Answer: (C)

34. HARD · NO CALCULATOR NEEDED

The exponential function $f(x) = ab^x$, where a and b are positive constants, satisfies $f(2) = 18$ and $f(5) = 486$. What is the value of $f(4)$?

1. Divide the two conditions: $\dfrac{f(5)}{f(2)} = b^3 = \dfrac{486}{18} = 27$, so $b = 3$.

2. Then $a \cdot 3^2 = 18$, so $a = 2$.

3. $f(4) = 2 \cdot 3^4 = 162$.

Answer: 162

35. HARD · NO CALCULATOR NEEDED

The size of a bacterial colony, in thousands of cells, is modeled by $V(t) = 500 \cdot 2^{t/2} \cdot 4^{t/3} \cdot 16^{-t/12}$, where t is the number of hours since the start of observation. The size of the colony is multiplied by a constant factor c every 6 hours. What is the value of c?

1. Use base 2: $4^{t/3} = 2^{2t/3}$ and $16^{-t/12} = 2^{-4t/12} = 2^{-t/3}$.

2. Add the exponents: $\frac{t}{2} + \frac{2t}{3} - \frac{t}{3} = \frac{5}{6}t$, so $V(t) = 500 \cdot 2^{\frac{5}{6}t}$.

3. Six hours later: $\frac{V(t+6)}{V(t)} = 2^{\frac{5}{6}\cdot 6} = 2^5 = 32$.

Answer: 32

PART B · SELF-ASSESSMENT — 19 PROBLEMS

36. MEDIUM · NO CALCULATOR NEEDED

What is the solution to $\sqrt{2x+7} = 5$?

1. Square both sides: $2x + 7 = 25$.

2. Subtract 7: $2x = 18$, so $x = 9$.

3. Check in the original equation: $\sqrt{2(9)+7} = \sqrt{25} = 5$.

Answer: 9

37. MEDIUM · CALCULATOR

The estimated population of a new coastal town t years after 2024 is modeled by

$$P(t) = 5000\,(1.15)^t\,.$$

According to the model, by approximately what factor does the population increase every 5 years?

(A) 1.15 (B) 1.75

(C) 2.01 ✓ (D) 2.99

1. $P(t+5) = 5000(1.15)^{t+5} = P(t) \cdot (1.15)^5$, so the factor per 5 years is 1.15^5 (the 5000 and t cancel).

2. Type `1.15^5`: it returns 2.0114.

3. The closest option is 2.01: option (C). The value 1.75 comes from wrongly using $1 + 0.15 \cdot 5$.

Answer: (C)

38. Medium · Calculator

A car is purchased for \$120,000. Its value depreciates according to

$$V(t) = 120000\,(0.88)^{t},$$

where t is the number of years since purchase. To the nearest thousand dollars, what is the value of the car after 4 years?

(A) \$61,000

(B) \$72,000 ✓

(C) \$84,000

(D) \$94,000

1. Substitute $t = 4$ into $V(t) = 120000(0.88)^{t}$.

2. Type `120000*0.88^4`: it returns 71,963 dollars.

3. To the nearest thousand dollars this is 72,000: option (B).

Answer: (B)

39. Medium · No calculator needed

The storage capacity of a battery facility is modeled by

$$E(t) = 1450\,(1.08)^{t},$$

where $E(t)$ is the capacity, in megawatt-hours, and t is the number of years after the facility opened, for $0 \le t \le 15$. Which is the best interpretation of 1450 in this context?

(A) The predicted capacity, in megawatt-hours, 15 years after the facility opened

(B) The estimated capacity, in megawatt-hours, when the facility opened ✓

(C) The annual percent increase in capacity

(D) The number of years it takes the capacity to double

1. In $E(t) = 1450(1.08)^{t}$, the time $t = 0$ is the moment the facility opened.

2. At $t = 0$: $E(0) = 1450(1.08)^{0} = 1450 \cdot 1 = 1450$.

3. So 1450 megawatt-hours is the capacity when the facility opened: option (B). The growth rate is the 8% hidden in 1.08, not the 1450.

Answer: (B)

40. Medium · Calculator

A population of bacteria in a lab culture is modeled by $P(t) = 300(1.5)^{t}$, where t is the time in hours. What is the population after 3 hours?

(A) 675

(B) 900

(C) 1,012.5 ✓

(D) 1,350

1. Substitute $t = 3$: $P(3) = 300(1.5)^{3}$.

2. Type `300*1.5^3`: it returns 1,012.5.

3. That is option (C). Do not compute $300 \cdot 1.5 \cdot 3 = 1{,}350$: that treats the growth as linear.

Answer: (C)

41. Medium · Calculator

The number of subscribers to an online tutoring platform grows according to $N(t) = 500(1.2)^t$, where t is measured in months since launch. Which of the following is closest to the number of subscribers after 5 months?

(A) 1,000

(B) 1,100

(C) 1,244 ✓

(D) 1,500

1. Substitute $t = 5$: $N(5) = 500(1.2)^5$.

2. Type `500*1.2^5`: it returns 1,244.16.

3. The closest option is 1,244: option (C).

Answer: (C)

42. Medium · Calculator

The amount of a medication remaining in a patient's bloodstream is modeled by $M(t) = 200(0.85)^t$, where t is the number of hours since the dose was taken and $M(t)$ is measured in milligrams. Which of the following is closest to the amount remaining after 6 hours?

(A) 65.5

(B) 75.5 ✓

(C) 85.5

(D) 95.5

1. Substitute $t = 6$: $M(6) = 200(0.85)^6$.

2. Type `200*0.85^6`: it returns 75.4299.

3. The closest option is 75.5: option (B).

Answer: (B)

43. Medium · No calculator needed

The population of a small town has grown at an average rate of 2.4% per year since 2010. There were approximately 60,000 residents in 2010. Which of the following functions best models the town's population P, in thousands of residents, t years after 2010?

(A) $P(t) = 60(1.024)^t$ ✓

(B) $P(t) = 60(2.4)^t$

(C) $P(t) = 1.024t + 60$

(D) $P(t) = 2.4t + 60$

1. A growth of 2.4% per year multiplies the population by $1 + 0.024 = 1.024$ each year.

2. The starting population is 60 thousand, so $P(t) = 60(1.024)^t$, option (A).

3. Check: after one year $P(1) = 60(1.024) = 61.44$ thousand, which is $60 + 1.44$ (that is 2.4% of 60).

Answer: (A)

44. Medium · No calculator needed

Each term in the sequence below is 6 times the previous term. What is the 9th term in the sequence?

$$3,\ 18,\ 108,\ 648,\ \ldots$$

(A) 648×9

(B) 3^9

(C) 3×6^8 ✓

(D) 6×3^8

1. Multiplying by 6 each time makes this a geometric sequence with first term 3 and ratio 6: the nth term is $3 \cdot 6^{n-1}$.

2. Check the 4th term: $3 \cdot 6^3 = 648$, matching the list.

3. The 9th term is $3 \cdot 6^8$, option (C).

Answer: (C)

45. Medium · No calculator needed

The first term of a geometric sequence is 2 and the fourth term is 54. What is the sum of the first five terms of the sequence?

1. The fourth term is $2r^3 = 54$, so $r^3 = 27$ and $r = 3$.

2. The first five terms are $2, 6, 18, 54, 162$.

3. Their sum is $2 + 6 + 18 + 54 + 162 = 242$.

Answer: 242

46. Hard · Desmos

The function $P(t) = \dfrac{120}{1 + 5(0.8)^t}$ models the number of active users of an app, in thousands, t weeks after launch, for $0 \le t \le 10$. Consider the following statements.

I. P is increasing on the interval $0 \le t \le 10$.

II. There is a time t in the interval at which $P(t) = 60$.

III. There is a time t in the interval at which $P(t) = 80$.

Which of the statements must be true?

(A) I only

(B) I and II only ✓

(C) II and III only

(D) I, II, and III

1. Type `y=120/(1+5*0.8^x){0<=x<=10}` (letting x stand for t). The curve rises from (0, 20) to (10, 78.0807), so P is increasing: statement I is true.

2. Add the line `y=60` and tap the intersection: (7.2126, 60). It lies inside the interval, so statement II is true.

3. Add the line `y=80`: it never meets the graph for $0 \le t \le 10$, because the highest point is $P(10) \approx 78.08$. Statement III is false.

4. Type `120/(1+5*0.8^10)`: it returns 78.08, which confirms the highest value. So I and II only: option (B).

Answer: (B)

47. HARD · NO CALCULATOR NEEDED

The value V, in dollars, of a piece of lab equipment t years after it was purchased is modeled by

$$V(t) = 2400\,(0.64)^{t/2.5}.$$

Which of the following is the best interpretation of the model?

(A) It falls 36% every 2.5 years. ✓

(B) It falls 36% every 5 years.

(C) It falls 64% every 2.5 years.

(D) It falls 2.5% every year.

1. Increasing t by 2.5 adds 1 to the exponent $\dfrac{t}{2.5}$, so V is multiplied by 0.64 every 2.5 years.
2. A factor of 0.64 means a fall of $1 - 0.64 = 0.36$, that is 36%, every 2.5 years.
3. Check the other options: every 5 years the factor is $0.64^2 = 0.4096$, a fall of about 59.04%, not 36%. The correct choice is (A).

Answer: (A)

48. HARD · NO CALCULATOR NEEDED

The number of bacteria in a colony t years after January 1, 2020 is modeled by $A(t) = 300 \cdot 4^{t/3}$. Which of the following gives the number of bacteria m months after January 1, 2023?

(A) $300 \cdot 2^{m/18}$

(B) $1200 \cdot 2^{m/36}$

(C) $900 \cdot 2^{m/18}$

(D) $1200 \cdot 2^{m/18}$ ✓

1. January 1, 2023 is 3 years after January 1, 2020, and m more months is $\dfrac{m}{12}$ more years, so $t = 3 + \dfrac{m}{12}$.
2. $A = 300 \cdot 4^{(3+m/12)/3} = 300 \cdot 4^{1+m/36} = 1200 \cdot 4^{m/36}$.
3. Convert the base: $4^{m/36} = 2^{2m/36} = 2^{m/18}$, so $A = 1200 \cdot 2^{m/18}$, option (D).
4. Check $m = 0$: $1200 \cdot 2^0 = 1200$, which equals $A(3) = 300 \cdot 4 = 1200$.

Answer: (D)

49. HARD · NO CALCULATOR NEEDED

$$T(t) = 24 + 66(0.85)^t$$

The function T models the temperature, in degrees Celsius, of a cup of tea t minutes after it is placed in a room. Which of the following is the best interpretation of the number 0.85 in this context?

(A) T falls 15% per minute

(B) the gap $T - 24$ falls 15% per minute ✓

(C) $T(1) = 0.85\,T(0)$

(D) T levels off at 85% of $T(0)$

1. Subtract the constant: $T(t) - 24 = 66(0.85)^t$. This is the gap between the tea and the room temperature.

2. Each extra minute multiplies the gap by 0.85, so the gap falls by 15% per minute: option (B).

3. Why not (C): $T(0) = 90$ and $T(1) = 80.1$, but $0.85 \cdot T(0) = 76.5 \neq T(1)$. The gap goes from 66 to $56.1 = 0.85 \cdot 66$.

Answer: (B)

50. HARD · CALCULATOR

The number of visitors to a national park increased by 25% from 2022 to 2023. From 2023 to 2024, the number of visitors decreased by 12%. If the number of visitors in 2024 was 9,020,000, what was the number of visitors in 2022?

1. Let x be the 2022 number. Successive changes multiply: $x \cdot 1.25 \cdot 0.88$.

2. Calculate the combined factor. Type `1.25*0.88`: it returns 1.1, so $1.1x = 9{,}020{,}000$.

3. Solve. Type `9020000/1.1`: it returns 8,200,000.

Answer: 8,200,000

51. MEDIUM · CALCULATOR

A jacket is discounted by 30%. A 5% sales tax is then applied to the discounted price. If the final price paid is \$44.10, what was the original price of the jacket, before the discount?

(A) \$42.00

(B) \$58.80

(C) \$60.00 ✓

(D) \$63.00

1. Let x be the original price. After a 30% discount the price is $0.7x$; the tax is applied to that: $0.7x \cdot 1.05$.

2. Combined factor: Type `0.7*1.05`: it returns 0.735, so $0.735x = 44.10$.

3. Solve: Type `44.10/0.735`: it returns 60 dollars, option (C).

Answer: (C)

52. Medium · Calculator

A laptop's price increased by 20% during a promotional period, then decreased by 15% after the promotion ended. If the final price is \$1,020, what was the price at the end of the promotional period, before the decrease?

(A) \$950

(B) \$1,000

(C) \$1,050

(D) \$1,200 ✓

1. The final price comes after the 15% decrease, so it is 85% of the price at the end of the promotion: $0.85p = 1{,}020$.
2. Solve: Type `1020/0.85`: it returns 1,200 dollars, option (D).
3. Check: Type `1200*0.85`: it returns 1,020. The original price (the trap in option 1,000) is found by type `1200/1.2`: it returns 1,000 dollars.

Answer: (D)

53. Hard · Calculator

The number of families visiting a city festival's family zone increased by 40% from Week 1 to Week 2, then decreased by 25% from Week 2 to Week 3. If Week 3 attendance was 8,400 families, how many more families attended in Week 2 than in Week 3?

1. Work backwards from Week 3 to Week 2. Week 3 is 75% of Week 2. Type `8400/0.75`: it returns 11,200 families in Week 2.
2. The question asks for the difference. Type `8400/0.75-8400`: it returns 2,800.
3. Check with Week 1. Type `8400/(1.4*0.75)`: it returns 8,000; type `1.4*8000`: it returns 11,200 in Week 2, and $11{,}200 - 8{,}400 = 2{,}800$.

Answer: 2,800

54. Medium · No calculator needed

Which of the following represents the result of decreasing the quantity y by 12%, where $y > 0$?

(A) $0.88y$ ✓

(B) $0.12y$

(C) $y - 12$

(D) $y - 0.12$

1. Decreasing by 12% removes $0.12y$: $y - 0.12y$.
2. Factor out y: $(1 - 0.12)y = 0.88y$, option (A).
3. Check with $y = 50$: a 12% decrease is 6, so $50 - 6 = 44$ and $0.88(50) = 44$.

Answer: (A)